USD/INR Rollover Watcher & Execution Engine

System Logic, Mathematical Calculations, & Operator Reference Guide

1. Core Concept: What is a Futures Rollover & How Do Orders Execute?

A currency futures contract has a fixed monthly expiry date. To maintain a foreign exchange hedge without holding cash delivery, an expiring position must be rolled forward into the next month. The automated engine executes both legs in **one unified clip**:

- **Near Leg (Oct FUT): SELL** at the Bid to close the expiring contract.
- **Far Leg (Nov FUT): BUY** at the Ask to establish the new forward contract.

CORE SIZING LOGIC: LOTS VS. QUANTITY

- **1 Lot** = **1,000 USD Units** (Exchange standard contract size for USDINR).
- **Your Total Holding: 4,000 Lots** = **4,000,000 Quantity** (40 Lakh USD).
- **Order Execution (1 Clip):** The bot is set to lots: 1. It sends **1,000 quantity in one go** (SELL 1,000 Oct / BUY 1,000 Nov).
- **Post-Trade Position:** Leaves 3,999 Lots in October. The app safely halts after 1 clip due to max_clips_per_day: 1.

2. Mathematical Calculation Breakdown (Every Metric on Your Screen)

Metric / Number	Formula & Derivation	Live Example	What It Means to the Trader
Near Leg (Oct FUT)	Marketable exit hits Bid: Near Bid = 94.9500 (Ask = 94.9550)	**94.9500** (10,000 resting)	The rupee price you receive when selling 1 lot of October futures.
Far Leg (Nov FUT)	Marketable entry lifts Ask: Far Ask = 95.4225 (Bid = 95.4025)	**95.4225** (100,000 resting)	The rupee price you pay when buying 1 lot of November futures.
Roll Cost (Spread)	Far Ask - Near Bid = 95.4225 - 94.9500	**0.4725 Rs**	The carry spread in rupees per dollar to switch contracts.
Reference Mid-Price	Near contract mid: (94.9500 + 94.9550) / 2	**94.9525 Rs**	The benchmark price used to convert rupee spread into basis points.
Cost in Basis Points (bps)	(Roll Cost / Mid) x 10,000 = (0.4725 / 94.9525) x 10,000	**49.8 bps** **(0.498%)**	Standard metric for roll expense (1 bps = 0.01% of underlying).
Cost in Rs per Lot	Roll Cost x Lot Size = 0.4725 x 1,000	**Rs 472.50** **per lot**	Total out-of-pocket carry cost to roll 1 lot (1,000 USD).
Configured Roll Limit (Target Ceiling)	(Limit bps / 10,000) x Mid = (47 / 10,000) x 94.9525	**47 bps** **(= 0.4462 Rs)**	The maximum spread you permit. Bot will reject anything higher.
Market Status & Gap	Current Cost - Limit = 49.8 bps - 47.0 bps	**WAITING** **2.8 bps too dear**	Market is 2.8 bps more expensive than your limit. App holds safely.
Target Execution Price	Near Bid + Limit Spread = 94.9500 + 0.4462	**95.3962** (or below)	If Far Ask drops to <= 95.3962 (or Oct Bid rises), the trade fires.

3. Execution Condition: When Does the System Trigger?

The system executes when: **Observed Roll Cost (bps) <= Configured Roll Limit (bps)**.

- If market is at 49.8 bps and limit is 47 bps: **Holds safely in WAITING state** (protects you from paying too much).
- If limit is updated to 50 bps (or market spread narrows to <= 47 bps): **Qualifies immediately** and fires the order.

4. Institutional Safety Gates & Built-in Capital Protections

Before sending any live order to the exchange, the engine automatically checks every safety gate:

Protection Gate	Configured Setting	Safety Guardrail & Purpose
Daily Clip Cap	max_clips_per_day: 1	Prevents runaway trading. After 1 lot completes, the engine stops automatically for the day.
Book Liquidity Gate	require_touch_size: true	Ensures resting book size at the touch exceeds clip size (1,000 units), preventing slippage.
Circuit Band Check	Exchange Floor / Ceiling	Validates that buy/sell limits are strictly within exchange upper & lower price limits.
Auto-Disarm Timer	arm_timeout_sec: 120	When armed in LIVE mode, it automatically disarms after 2 minutes if the spread does not qualify.
Two-Leg Fail-Safe	auto_unwind: false	If Leg 1 fills and Leg 2 experiences network disruption, the bot halts immediately without guessing.
Margin Gate	Pre-trade check	Verifies available free margin (Rs 25.70 Lakh) easily covers the required margin (Rs 4,572 per lot).

5. Operator Step-by-Step Workflow (How to Trade via GUI)

Step	Action in Application Window (dist/roll_app.exe)	Expected Screen Response
Step 1	**Launch roll_app.exe** Verify October & November contract tokens and live market prices.	Near Bid, Far Ask, and current Roll Cost (e.g. 0.4725 / 49.8 bps) update in real time.
Step 2	**Set / Verify Roll Limit** Enter target bps (e.g. 47 bps to wait, or 50 bps for immediate fill) and click **Set**.	The limit card updates with the new rupee threshold and recalculates the difference in bps.
Step 3	**Switch from DRY RUN to LIVE** Click the **DRY RUN** toggle at top. A safety preflight dialog opens. Type LIVE in the box and click Confirm.	The mode indicator changes to a bold red **LIVE** badge. System is authorised to place orders.
Step 4	**Click ARM** System arms for 120 seconds.	As soon as Cost <= Limit, it sells 1,000 Oct / buys 1,000 Nov, logs fills, and halts safely.

6. Trader Quick Reference & FAQ

- **Will the bot accidentally roll all 4,000 lots at once?**
  No. Order size is strictly controlled by lots: 1 (1,000 quantity = 1 lot). With max_clips_per_day: 1, it executes only 1 lot and stops.
- **Why does the order payload say 1,000 quantity instead of 1 lot?**
  Choice Broking exchange API requires order quantity in base units (1 lot = 1,000 units). The app lets you think in lots and translates to units automatically.
- **What does '2.8 bps too dear' mean?**
  'Too dear' means too expensive. The live spread (49.8 bps) is 2.8 bps higher than your limit (47.0 bps). The app holds until the market reaches your price.